

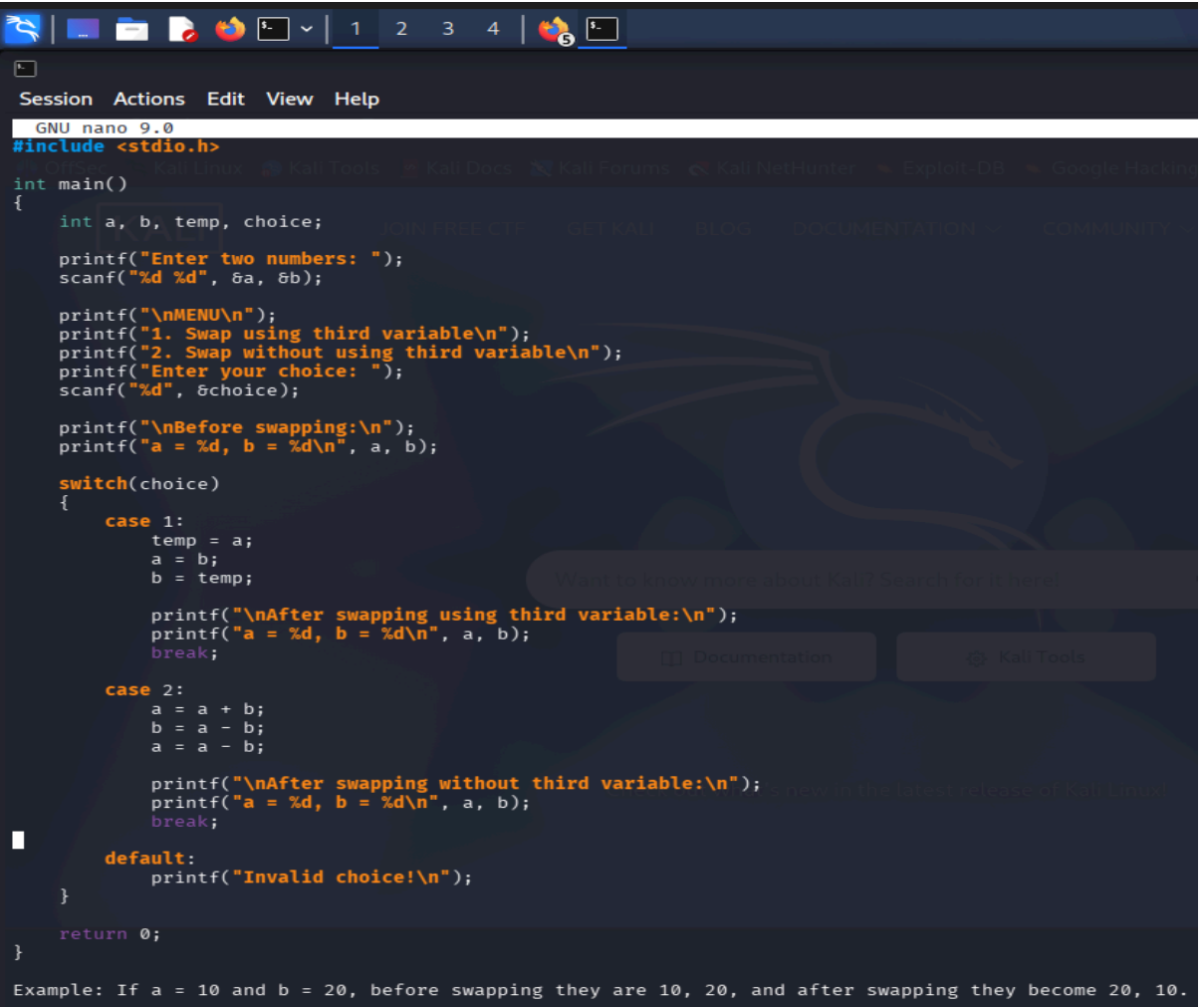
Experiment-9

Q1. Write a menu driven C program to input two numbers and swap them using a third variable and without using third variable. Also display the values before and after swapping.

1. Swap Two Numbers

The menu gives two methods:

1. Swap using a third variable
2. Swap without using a third variable



```
GNU nano 9.0
#include <stdio.h>

int main()
{
    int a, b, temp, choice;

    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);

    printf("\nMENU\n");
    printf("1. Swap using third variable\n");
    printf("2. Swap without using third variable\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    printf("\nBefore swapping:\n");
    printf("a = %d, b = %d\n", a, b);

    switch(choice)
    {
        case 1:
            temp = a;
            a = b;
            b = temp;

            printf("\nAfter swapping using third variable:\n");
            printf("a = %d, b = %d\n", a, b);
            break;

        case 2:
            a = a + b;
            b = a - b;
            a = a - b;

            printf("\nAfter swapping without third variable:\n");
            printf("a = %d, b = %d\n", a, b);
            break;

        default:
            printf("Invalid choice!\n");
    }

    return 0;
}
```

Example: If a = 10 and b = 20, before swapping they are 10, 20, and after swapping they become 20, 10.

Q2. Write a menu driven C program to find the sum of the first n natural numbers and the sum of squares of the first n natural numbers.

2. Sum of Natural Numbers and Their Squares

The menu gives:

1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers

```
Session Actions Edit View Help
GNU nano 9.0

#include <stdio.h>

int main()
{
    int n, choice, i;
    int sum = 0;

    printf("Enter the value of n: ");
    scanf("%d", &n);

    printf("\nMENU\n");
    printf("1. Sum of first n natural numbers\n");
    printf("2. Sum of squares of first n natural numbers\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch(choice)
    {
        case 1:
            for(i = 1; i <= n; i++)
            {
                sum = sum + i;
            }
            printf("Sum of first %d natural numbers = %d\n", n, sum);
            break;

        case 2:
            for(i = 1; i <= n; i++)
            {
                sum = sum + (i * i);
            }
            printf("Sum of squares of first %d natural numbers = %d\n",
                n, sum);
            break;

        default:
            printf("Invalid choice!\n");
    }
}
```

For example, if $n = 5$:

Sum = $1 + 2 + 3 + 4 + 5 = 15$

Sum of squares = $1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 55$

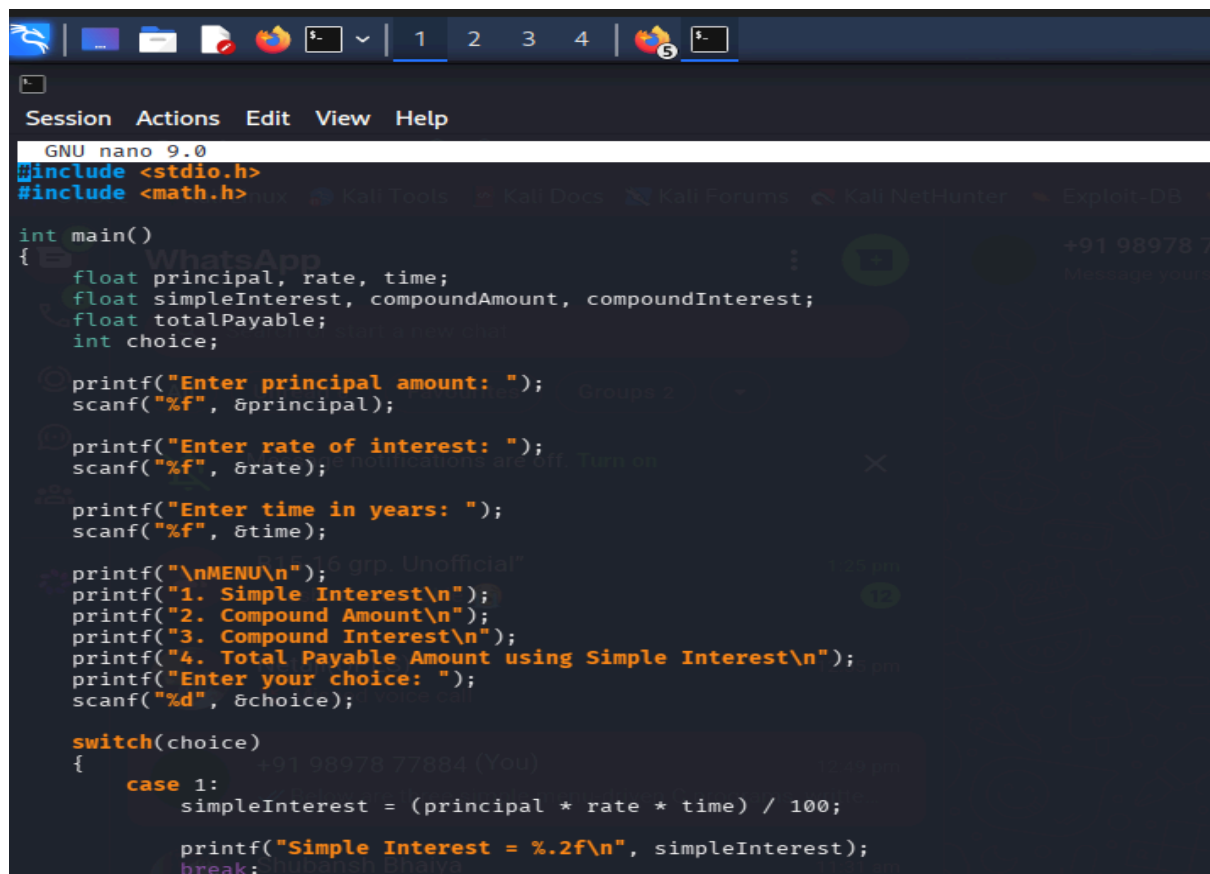
Q3, Write a menu driven C program to calculate simple interest, compound amount, and total payable amount for given principal, rate and time.

3. Simple Interest and Compound Interest

The menu calculates:

1. Simple Interest
2. Compound Amount
3. Compound Interest
4. Total Payable Amount

Here, compound interest is calculated assuming annual compounding.

A screenshot of a Linux terminal window showing a C program being edited in the nano text editor. The program is a menu-driven application for calculating interest. It includes headers for stdio.h and math.h. The main function prompts the user for principal, rate, and time, then displays a menu with four options: Simple Interest, Compound Amount, Compound Interest, and Total Payable Amount using Simple Interest. The code for the first case (Simple Interest) is shown, using the formula (principal * rate * time) / 100. The editor's status bar at the bottom indicates 'GNU nano 9.0'.

```
GNU nano 9.0
#include <stdio.h>
#include <math.h>

int main()
{
    float principal, rate, time;
    float simpleInterest, compoundAmount, compoundInterest;
    float totalPayable;
    int choice;

    printf("Enter principal amount: ");
    scanf("%f", &principal);

    printf("Enter rate of interest: ");
    scanf("%f", &rate);

    printf("Enter time in years: ");
    scanf("%f", &time);

    printf("\nMENU\n");
    printf("1. Simple Interest\n");
    printf("2. Compound Amount\n");
    printf("3. Compound Interest\n");
    printf("4. Total Payable Amount using Simple Interest\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch(choice)
    {
        case 1:
            simpleInterest = (principal * rate * time) / 100;
            printf("Simple Interest = %.2f\n", simpleInterest);
            break;
```

```

    case 2:
        compoundAmount =
            principal * pow((1 + rate / 100), time);
        printf("Compound Amount = %.2f\n", compoundAmount);
        break;

    case 3:
        compoundAmount =
            principal * pow((1 + rate / 100), time);

        compoundInterest = compoundAmount - principal;

        printf("Compound Interest = %.2f\n", compoundInterest);
        break;

    case 4:
        simpleInterest = (principal * rate * time) / 100;
        totalPayable = principal + simpleInterest;

        printf("Total Payable Amount = %.2f\n", totalPayable);
        break;

    default:
        printf("Invalid choice!\n");
}

return 0;
}

```

Formulas used

Simple Interest:

$$SI = \frac{P \times R \times T}{100}$$

Compound Amount:

$$A = P \left(1 + \frac{R}{100} \right)^T$$

Compound Interest:

$$CI = A - P$$

Total payable amount with simple interest:

$$\text{Total Payable} = P + SI$$

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